On Query-Side Compression for Online-Efficient Dense Retrieval: Failure Modes, Pooling Effects, and ANN Tradeoffs

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Abstract

Dense retrieval often places the query encoder on the online critical path while document encoding is amortized offline. This asymmetry motivates query-side compression, but it is unclear how much retrieval quality survives when only the query tower is aggressively reduced. We study this question with a frozen zero-shot `BAAI/bge-base-en-v1.5` teacher, asymmetric 4-layer and 2-layer students, and a controlled MS MARCO-derived benchmark with explicit latency measurement. Naive truncation causes severe retrieval collapse. Among low-cost interventions, query-side mean pooling is the first clear positive change, whereas a query-side projection head hurts and lightweight distillation is nearly neutral. A later systems comparison shows that IVF, but not HNSW, provides a plausible search-speed tradeoff relative to exact flat search. The resulting picture is narrow but defensible: shallow asymmetric query compression is brittle under simple truncation, and internal design choices matter before strong efficiency claims can be made.

1 Introduction

Dense retrieval is attractive when lexical matching is too brittle, but its deployment cost is not symmetric. In many production settings, document embeddings can be computed offline and indexed once, whereas query encoding lies directly on the online critical path. This difference makes query-side efficiency more operationally important than overall encoder symmetry. The relevant question is therefore not only how to improve a dual encoder in the style of DPR and related bi-encoder systems [5, 8], but how much retrieval quality survives when the document tower remains strong and only the query tower is compressed.

This paper studies that question in a deliberately narrow form. We fix a strong zero-shot teacher, build asymmetric students by truncating the query tower alone, and then test a small sequence of low-cost interventions: pooling, a projection head motivated by contrastive representation learning [2], lightweight distillation [3], and a later approximate-nearest-neighbor (ANN) systems comparison implemented with FAISS-style indices [4, 6]. The intent is not to present a new state-of-the-art retriever. The intent is to isolate failure modes and first-order tradeoffs in online-efficient dense retrieval.

The resulting empirical picture is sharper than the original hypothesis suggested. A zero-shot `BAAI/bge-base-en-v1.5` teacher remains extremely strong on the prepared MS MARCO-derived benchmark. Naive 4-layer and 2-layer query truncation causes catastrophic retrieval collapse. Query-side mean pooling becomes the first clear positive intervention on the 4-layer student, but the best student still remains far below the teacher. A small query-side projection head hurts retrieval quality, and the tested lightweight distillation objective does not materially improve the tradeoff. On the systems side, IVF is the only ANN variant that preserves enough quality to be worth discussion; HNSW is faster but too lossy under this setup.

These observations support a short, evidence-driven conclusion. Query-side compression is possible to study cleanly in an asymmetric setting, but shallow compression is highly brittle under simple truncation. Internal representation choices matter before broader efficiency claims can be made, and BEIR-style transfer checks [9] are useful only after the in-domain tradeoff is understood. The main value of the paper is therefore a controlled map of what fails first, what helps first, and which systems extension is plausible once the encoder story is stabilized.

2 Related Work

Dense retrieval as bi-encoding. Dense passage retrieval is usually cast as a dual-encoder problem in which queries and passages are embedded independently and scored with an inner product or cosine sim-

ilarity. DPR established that this simple formulation can replace sparse first-stage retrieval in open-domain QA, and Sentence-BERT showed that siamese and triplet-style text encoders make single-vector semantic search practical at scale [5, 8]. Our study uses that standard bi-encoder setup and asks how much quality remains when only the query side is compressed.

Contrastive training and representation heads. In-batch contrastive learning is the default optimization recipe for dense retrievers, and SimCLR is the canonical reference for normalized contrastive objectives and projection heads [2]. Dense retrieval systems such as RocketQA show that performance is sensitive to negatives, batching, and training/inference mismatch, while recent distillation work such as PairDistill continues to refine teacher-student supervision for retrieval [7, 3]. These lines of work motivate our pooling, projection-head, and lightweight distillation ablations, but they are broader than the narrow asymmetric setting studied here.

Embedding teachers and zero-shot evaluation. Strong embedding models now serve as practical frozen references for retrieval studies. The official `BAAI/bge-base-en-v1.5` model card documents the retrieval instruction and CLS-based embedding usage that we follow for the teacher reference, and it reports strong zero-shot retrieval performance on standard benchmarks [1]. For transfer evaluation, BEIR remains the standard heterogeneous zero-shot benchmark and is well aligned with our emphasis on generalization rather than leaderboard fitting [9].

ANN indexing. At query time, dense retrieval systems rely on exact or approximate nearest-neighbor search over a fixed vector index. Faiss is the standard library for this layer, and its flat, IVF, and HNSW families provide the canonical exact and approximate baselines for large-scale vector search [4, 6]. In this paper, ANN is treated as a systems comparison layered on top of the retrieval model, not as a new retrieval objective.

3 Method

We study an asymmetric dense retriever with a full-depth document encoder and a compressed query encoder. Let q denote a query and d denote a document. The query tower is truncated to depth $L_q \in \{2, 4\}$, while the document tower remains full depth and is frozen for all student experiments. For a text input x, each tower produces hidden states $H_r(x) \in \mathbb{R}^{T \times h}$

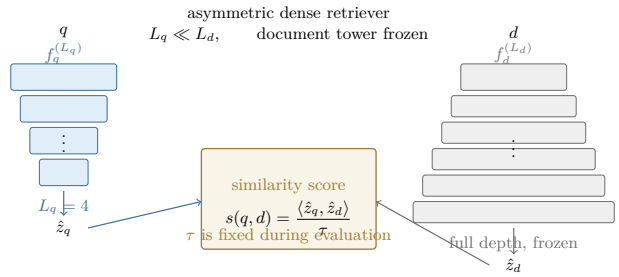

Figure 1: Asymmetric dense retrieval setup used throughout the student experiments. The query tower is truncated to depth L_q, while the document tower remains full-depth and frozen. Similarity is computed by scaled inner product over normalized embeddings.

with attention mask $m(x)$, where $r \in \{q, d\}$ indexes the query or document encoder.

We pool hidden states with a low-cost operator $p_\rho(\cdot)$, where $\rho \in \{\text{cls}, \text{mean}\}$. The pooled vector is optionally transformed by a query-side projection head g_ϕ and then ℓ_2-normalized:

$$z_q(q) = \text{norm}\big(g_\phi\big(p_{\rho_q}(H_q(q), m(q))\big)\big)\,, \tag{1}$$

$$z_d(d) = \text{norm}(p_{\rho_d}(H_d(d), m(d)))\,. \tag{2}$$

In the symmetric teacher, the same backbone is used on both sides; in the asymmetric students, only the query encoder is truncated and only the query side may use a projection head.

Retrieval scores are scaled inner products:

$$s(q, d) = \alpha\, z_q(q)^\top z_d(d), \qquad \alpha = \exp(\tau), \tag{3}$$

where τ is a learned log-temperature. This matches the study setup: normalized embeddings, inner-product similarity, and a learned scalar scale parameter.

Training uses an in-batch contrastive objective over positive query–document pairs. For a batch $\{(q_i, d_i^+)\}_{i=1}^B$, the logits are

$$\ell_{ij} = s(q_i, d_j^+), \tag{4}$$

and the loss is

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(\ell_{ii})}{\sum_{j=1}^{B} \exp(\ell_{ij})}. \tag{5}$$

This is the only training signal for the baseline student and teacher fine-tuning runs.

When distillation is enabled, the student is regularized to match a frozen teacher similarity distribution over the same batch. Let ℓ_{ij}^S and ℓ_{ij}^T denote student

and teacher logits. With distillation temperature T, the per-query distributions are

$$p_i^S = \mathrm{softmax}\big(\ell_{i\cdot}^S / T\big)\,, \qquad p_i^T = \mathrm{softmax}\big(\ell_{i\cdot}^T / T\big)\,, \tag{6}$$

and the distillation term is

$$\mathcal{L}_{\mathrm{KL}} = \frac{T^2}{B} \sum_{i=1}^{B} \mathrm{KL}\big(p_i^T \,\|\, p_i^S\big)\,. \tag{7}$$

The total objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{ctr}} + \lambda_{\mathrm{distill}} \mathcal{L}_{\mathrm{KL}}, \tag{8}$$

with $\lambda_{\mathrm{distill}} = 0$ for non-distilled runs.

Online cost is decomposed into query encoding and approximate or exact search:

$$T_{\mathrm{e2e}}(q) = T_{\mathrm{encode}}(q) + T_{\mathrm{search}}(q). \tag{9}$$

In our measurements, T_{encode} is batch-size-1 query encoding latency and T_{search} is FAISS search latency against a fixed corpus index. We report p50 and p95 over a fixed query sample and separate the corpus indexing cost from query-time cost so the systems tradeoff is explicit.

4 Experimental Setup

The benchmark uses an MS MARCO-derived slice prepared with seed 13. The resulting training and retrieval files contain 100,000 training triples, 351,695 corpus passages, 55,578 development queries, and 59,273 qrels. Training and evaluation text are truncated to 256 tokens, and all query texts use the retrieval instruction prefix `Represent this sentence for searching relevant passages:` while passages use no prefix.

We report a frozen zero-shot teacher reference based on `BAAI/bge-base-en-v1.5` with CLS pooling. The main student family is the asymmetric 4-layer query encoder with mean pooling on the query side and CLS pooling on the passage side, which is the best-performing shallow student in the present study. Additional ablations include the raw 4-layer and 2-layer CLS students, a query-side projection-head variant, and a lightweight KL distillation variant. The teacher fine-tuning ablation is retained only as a comparison point; it does not replace the frozen zero-shot reference.

Training uses AdamW with batch size 16, weight decay 0.01, warmup ratio 0.1, and fixed learning rates for each run family. The teacher and student runs use the same tokenizer and the same MS MARCO-derived triples, with validation performed on a deterministic

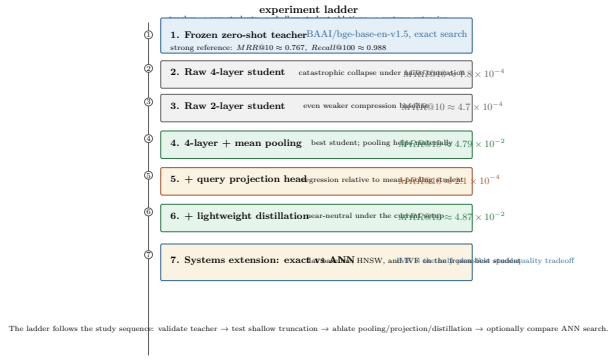

Figure 2: Experiment ladder for the study: frozen teacher, raw shallow students, low-cost ablations, and then the optional ANN systems extension.

5,000-example slice of the training triples. Checkpoint selection is based on retrieval probe `MRR@10` when a probe is enabled, which keeps the selection criterion aligned with retrieval quality rather than only validation loss.

Retrieval evaluation is performed on the full prepared corpus and development query set with FAISS exact inner-product search for the main tables. We report MRR@10 and Recall@100 together with corpus encoding time, index build time, query encoding latency, search latency, and end-to-end online latency. Latency is measured with batch size 1 after a short warmup, using p50 and p95 over a fixed 100-query sample. Query-side memory is reported from the measured latency runs. The main quantitative summaries consist of a retrieval table, a latency table, an ANN table, and one latency–quality tradeoff plot.

For robustness, we also ran a BEIR transfer check on SciFact using the strongest shallow student. That transfer result is reported as an auxiliary sanity check rather than a primary target. The ANN extension compares exact FAISS against HNSW and IVF on the strongest shallow student, using the same corpus/query slice and the same retrieval metrics so that speed and quality are directly comparable.

5 Results

The central result is that the frozen zero-shot teacher remains the strongest point in the study, while naive shallow compression collapses retrieval quality unless the query tower is handled carefully. Table 1 summarizes the quality tradeoffs, Table 2 summarizes online behavior, and Table 3 reports the exact-versus-ANN comparison.

5.1 Retrieval quality

The frozen teacher reference, `BAAI/bge-base-en-v1.5` with CLS pooling, achieves MRR@10 = 0.766997 and Recall@100 = 0.988237 on the prepared MS MARCO slice. The conservative fine-tuned teacher remains slightly worse at 0.740860/0.983169, so the paper uses the zero-shot model as the teacher reference rather than the fine-tuned ablation.

The shallow student results show a large quality cliff. The raw 4-layer student reaches only 0.000776/0.016888, and the 2-layer student is even weaker at 0.000465/0.010698. This is the clearest evidence that naive query truncation is not a viable compression strategy under the current setup.

Among the student ablations, query-side mean pooling is the first clearly positive intervention. The 4-layer student with mean pooling improves to 0.047922/0.249231, which is still far below the teacher but is orders of magnitude better than the raw shallow student. The query projection head is a regression: it drops to 0.000208/0.004516. Lightweight distillation is near-neutral relative to the pooling-only student, nudging MRR@10 to 0.048725 while slightly lowering Recall@100 to 0.246582.

5.2 Latency and systems behavior

The latency measurements in Table 2 show that the student-side query encoder can be made materially cheaper, but exact end-to-end retrieval is still dominated by corpus search. The best 4-layer mean-pooling student reduces query p50 latency to 1.819 ms from 3.089 ms for the frozen teacher, yet end-to-end p50 latency remains similar at 25.243 ms versus 23.567 ms. The 2-layer student is faster than the teacher at query time, but its retrieval quality is too poor to justify the stronger compression. The projection-head and distillation variants do not meaningfully change the systems picture.

5.3 ANN tradeoff

Table 3 compares exact FAISS search against two approximate indices on the frozen best student. `HNSW` yields the fastest mean search time at 0.038 ms, but its retrieval quality drops sharply to 0.032009/0.147751, which is too large a loss for the present study. `IVF` is the only ANN variant with a defensible tradeoff: it preserves much more of the exact-search quality at 0.044107/0.223453 while reducing mean search time to 1.151 ms from 3.579 ms for exact search. For this benchmark, `IVF` is the only approximate index worth mentioning as a plausible systems compromise.

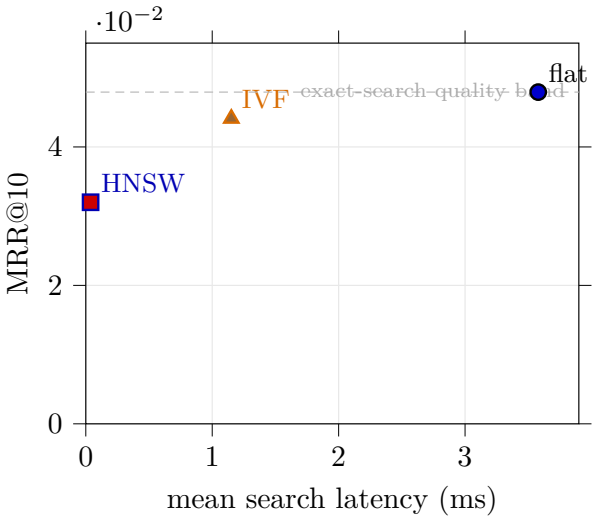

Figure 3: ANN search-latency versus quality on the frozen best student. HNSW is fastest but too lossy; IVF retains much more of the exact-search quality while still reducing mean search latency.

Table 1: Retrieval quality on the prepared MS MARCO slice. The zero-shot teacher is the frozen reference model. The 4-layer mean-pooling student is the strongest student ablation, but it still trails the teacher by a wide margin.

Model	Family	MRR@10	Recall@100	Notes
Teacher Zero-Shot	teacher	0.766997	0.988237	Frozen reference teacher (`BAAI/bge-base-en-v1.5`, CLS)
Teacher Fine-Tune	teacher_ablation	0.740860	0.983169	Conservative fine-tuned teacher, still below zero-shot
Student Q4 CLS	student	0.000776	0.016888	Raw 4-layer query truncation collapses quality
Student Q2 CLS	student	0.000465	0.010698	Raw 2-layer query truncation is weaker still
Student Q4 Mean	student_best	0.047922	0.249231	Best student: query-side mean pooling recovers signal
Student Q4 Mean + Proj	student_ablation	0.000208	0.004516	Projection head regression
Student Q4 Mean + Distill	student_ablation	0.048725	0.246582	Lightweight distillation is near-neutral

5.4 Transfer check

As a small robustness check, the frozen best student transfers to SciFact with MRR@10 = 0.040016 and Recall@100 = 0.326111 on the BEIR evaluation run. This is far below the MS MARCO teacher reference, but it confirms that the student remains usable outside the in-domain benchmark.

6 Discussion

The dominant pattern in these experiments is not a clean compression success story. It is a brittleness story. Once query depth is truncated naively, most of the teacher–student gap appears immediately, and that gap is large enough that subsequent low-cost interventions operate in a damaged regime rather

Table 2: Online behavior on the prepared benchmark. Query latency changes substantially across query encoders, but end-to-end latency remains dominated by exact corpus search. Peak memory is the measured query-side peak allocation for the benchmark run, reported in MiB for readability.

Model	Family	Query p50 (ms)	Query p95 (ms)	End-to-end p50 (ms)	End-to-end p95 (ms)	Peak memory (MiB)
Teacher Zero-Shot	teacher	3.089	3.199	23.567	23.821	429.25
Teacher Fine-Tune	teacher_ablation	3.075	3.213	23.837	24.243	1690.83
Student Q4 CLS	student	7.171	8.206	31.239	32.680	0.06
Student Q2 CLS	student	4.107	4.478	24.800	25.405	0.06
Student Q4 Mean	student_best	1.819	1.991	25.243	26.183	1662.73
Student Q4 Mean + Proj	student_ablation	1.848	1.957	24.822	25.105	1669.84
Student Q4 Mean + Distill	student_ablation	1.824	1.999	23.172	23.902	2079.23

Table 3: Exact FAISS versus approximate indices on the frozen best student. All rows use the same 55,578-query benchmark slice. `IVF` is the only approximate index that preserves most of the exact-search quality while still reducing mean search latency.

Index	MRR@10	Recall@100	Build (s)	Search (s)	Mean search (ms)	Queries	Notes
Flat	0.047922	0.249231	0.450	198.904	3.579	55,578	Exact FAISS baseline
HNSW	0.032009	0.147751	32.043	2.108	0.038	55,578	Fastest search, but large quality drop
IVF	0.044107	0.223453	3.285	63.974	1.151	55,578	Moderate speedup with smaller quality loss

than near the teacher frontier.

This behavior matters for two reasons. First, it clarifies that online-efficient dense retrieval should not be framed only as a latency problem. It is a representation problem whose failure modes appear before the search backend becomes the main bottleneck. Second, it explains why query-side pooling emerges as the first useful intervention: once the query tower is shallow, preserving token information in the final sentence representation becomes more important than simply inheriting the teacher's default end-of-tower behavior.

The negative projection-head result is also informative. In richer contrastive settings, a projection head can reshape the representation space without harming the underlying encoder. Here, the same idea appears to introduce extra mismatch against a frozen full-depth document tower. The lightweight distillation result points in a similar direction. Distillation helps only marginally when the student base representation is already too weak, suggesting that stronger student architecture or richer teacher signals may be needed before KL-based imitation can close the gap.

The ANN comparison should also be interpreted carefully. It does not rescue a weak student. Instead, it shows what can still be gained once the encoder is fixed. HNSW offers the most aggressive search-speed improvement but loses too much retrieval quality. IVF is the only ANN variant that moves the latency side of the tradeoff without rewriting the paper's qualitative conclusion.

7 Limitations

This study is intentionally narrow. The main benchmark is a reduced MS MARCO-derived slice rather than the full passage ranking task, so the reported tradeoffs should be read as controlled experimental evidence rather than a final production ranking claim. Only one teacher backbone is used, and the teacher remains strongest in its zero-shot form. The student space is also small: query truncation, pooling, a single projection-head design, and one lightweight distillation formulation. Other compression strategies, stronger distillation schedules, or better student initialization schemes may behave differently.

The transfer evidence is similarly limited. SciFact is useful as a robustness check, but one BEIR dataset does not establish broad cross-domain behavior. Finally, the ANN study is secondary to the core encoder question. It shows that systems-side approximations can matter once the encoder is frozen, but it does not change the central conclusion that the main difficulty lies in shallow query representation quality.

8 Conclusion

This paper studied online-efficient dense retrieval through the specific lens of asymmetric query compression. On a fixed MS MARCO-derived benchmark, a frozen zero-shot `BAAI/bge-base-en-v1.5` model remained the strongest teacher reference. Naive 4-layer and 2-layer query truncation caused catastrophic quality collapse. Query-side mean pooling yielded the first clear recovery for a shallow student, while the tested projection head hurt and lightweight distillation remained close to neutral. In the later systems comparison, IVF emerged as the only ANN option that preserved enough retrieval quality to be a plausible tradeoff against exact flat search.

The main conclusion is therefore narrow and practical. Query-side compression is not well approximated by simple truncation, and the first useful gains come from internal representation choices rather than from increasingly elaborate systems tricks. The remaining gap to the teacher is still large, which makes stronger distillation, better student architectures, or more careful shallow-query design the natural next research directions.

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